www.RocketRevise.com

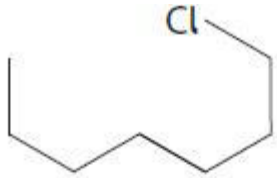
Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

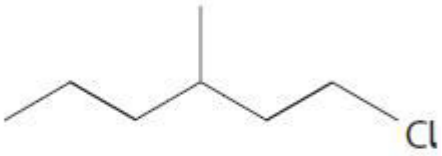
Which of these isomers has the **highest** boiling temperature?

☐ **A**

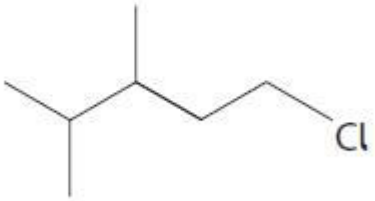


RocketRevise

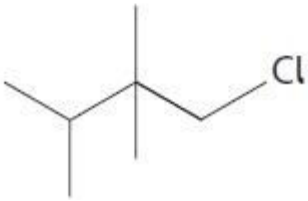
☐ **B**



☐ **C**



☐ **D**



Use this space for any rough working. Anything you write in this space will gain no credit.

RocketRevise

(Total for question = 1 mark)

www.RocketRevise.com

www.RocketRevise.com

Q2.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about hydrogen bonding.

(a) Which property is **not** due to hydrogen bonding?

(1)

- ☒ **A** ice has a lower density than water at 0°C
- ☒ **B** hydrogen fluoride has a higher boiling temperature than hydrogen chloride
- ☒ **C** H—H bond enthalpy is greater than Si—H bond enthalpy
- ☒ **D** alcohols are less volatile than alkanes with a similar molar mass

RocketRevise

(b) Which diagram best represents a hydrogen bond between two water molecules?

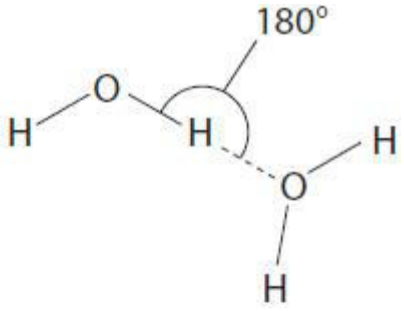
(1)

RocketRevise

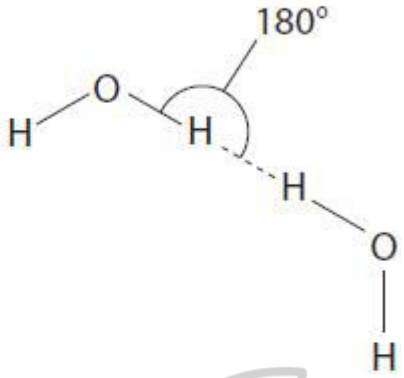
www.RocketRevise.com

www.RocketRevise.com

☒ **A**

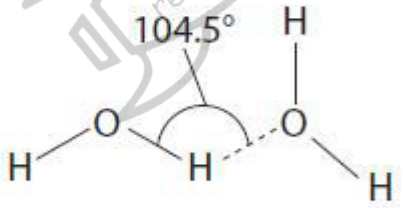


☒ **B**

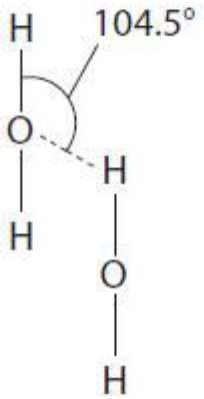


RocketRevise

☒ **C**



☒ **D**



RocketRevise

(Total for question = 2 marks)

Q3.

What are the strongest interactions **between** molecules in solid hydrogen iodide, HI?

www.RocketRevise.com

www.RocketRevise.com

(1)

- ☒ **A** covalent bonds
- ☒ **B** hydrogen bonds
- ☒ **C** ionic bonds
- ☒ **D** London forces

(Total for question = 1 mark)

Q4.

RocketRevise

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The skeletal formulae of three isomers are shown.



Which series shows the correct order of **increasing** boiling temperatures?

- ☒ **A** Y, X, Z
- ☒ **B** X, Y, Z
- ☒ **C** Y, Z, X
- ☒ **D** X, Z, Y

(Total for question = 1 mark)

Q5.

RocketRevise

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of the following statements about water is **not** due to hydrogen bonding?

- ☒ **A** water has a less open structure than ice
- ☒ **B** ice cubes float in a glass of iced water
- ☒ **C** when water freezes its volume increases
- ☒ **D** water is a good solvent for ionic compounds

(Total for question = 1 mark)

www.RocketRevise.com

www.RocketRevise.com

Q6.

Enthalpy changes of formation are often difficult to determine directly.

Some enthalpy data are shown.

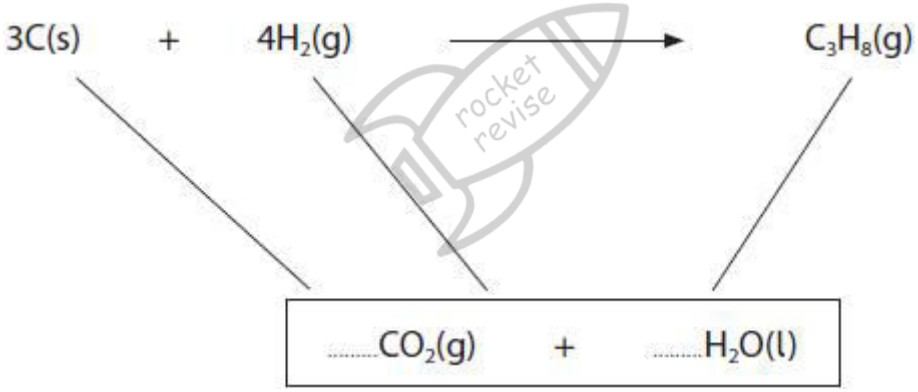
Compound	Standard enthalpy change of formation, $\Delta_f H^{\ominus}$ / kJ mol^{-1}
$H_2O(l)$	−285.8
$CO_2(g)$	−393.5

RocketRevise

Standard enthalpy change of combustion of propane $(\Delta_c H^{\ominus}(C_3H_8)) = -2219\,kJ\,mol^{-1}$.

(a) (i) Add arrowheads and stoichiometric coefficients to the Hess's Law diagram.

(1)



(ii) Use the data at the start of the question and your Hess's Law diagram to calculate the standard enthalpy change of formation of propane.
Include a sign and units in your answer.

(2)

RocketRevise

www.RocketRevise.com

www.RocketRevise.com

(b) The values for the boiling temperatures and the standard enthalpies of combustion of a series of straight-chain alkanes
are shown in the table.

Alkane	Boiling temperature / °C	Standard enthalpy change of combustion, $\Delta_c H^{\ominus}$ / kJ mol^{-1}	Increase in $\Delta_c H^{\ominus}$ / kJ mol^{-1}
C_2H_6	–88.5	–1560	–
C_3H_8	–42.0	–2219	659
C_4H_{10}	–0.5	–2877	658
C_5H_{12}	36.1	–3509	632
C_6H_{14}	68.8	–4163	654
C_7H_{16}	98.4	–4817	654

RocketRevise

(i) Explain why the increases in the values of $\Delta_c H^{\ominus}$ are similar.

(2)

.....

.....

.....

.....

.....

.....

.....

(ii) The increase in the value of $\Delta_c H^{\ominus}$ from butane to pentane is smaller than any other increase. Suggest an explanation for this.

(2)

.....

RocketRevise

.....

.....

.....

.....

.....

(iii) Explain, with reference to their intermolecular forces, why the boiling temperatures of **alkanes** increase as the
number of carbon atoms increases. A detailed description of the intermolecular forces is not required.

(3)

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

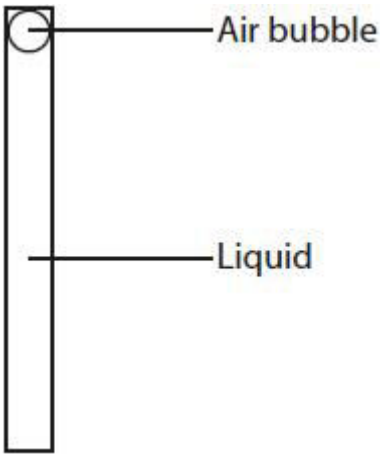
RocketRevise

(Total for question = 10 marks)

Q7.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The viscosities of four liquid organic compounds are compared by placing the liquids in separate identical tubes, each with an air bubble at the top.



RocketRevise

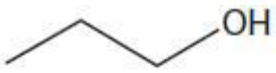
The tubes are inverted, and the times measured for the air bubble to travel the length of the tube.

For which compound does the air bubble take the longest time?

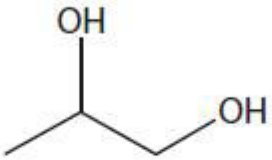
www.RocketRevise.com

www.RocketRevise.com

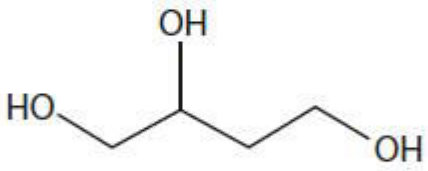
☐ **A**



☐ **B**

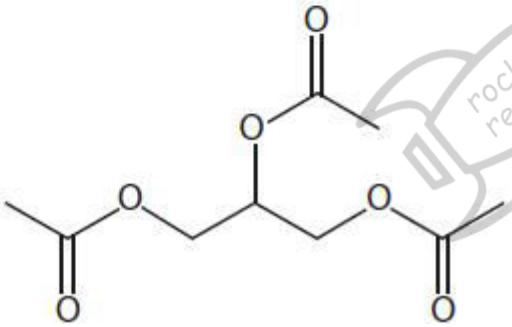


☐ **C**



RocketRevise

☐ **D**



(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these pure compounds has hydrogen bonding in the liquid state?

☐ **A** 1,1,1-trichloroethane, CH_3CCl_3

☐ **B** trimethylamine, $(CH_3)_3N$

☐ **C** hydrogen fluoride, HF

☐ **D** hydrogen sulfide, H_2S

(Total for question = 1 mark)

www.RocketRevise.com

www.RocketRevise.com

Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these isomers has the **highest** boiling temperature?

☐ **A**



RocketRevise

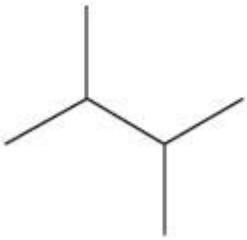
☐ **B**



☐ **C**



☐ **D**



RocketRevise

(Total for question = 1 mark)

Q10.

Which row shows the hydrogen halides in order of **increasing** boiling temperature?

www.RocketRevise.com

www.RocketRevise.com

	Lowest →			Highest
☐ A	HF	HCl	HBr	HI
☐ B	HI	HBr	HCl	HF
☐ C	HCl	HBr	HI	HF
☐ D	HF	HI	HBr	HCl

(Total for question = 1 mark)

RocketRevise

Q11.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these isomers with the formula C_8H_{18} has the lowest boiling temperature?

☐ **A** $CH_3CH(CH_3)CH_2CH_2CH(CH_3)CH_3$

☐ **B** $CH_3CH(CH_3)CH_2CH_2CH_2CH_2CH_3$

☐ **C** $CH_3C(CH_3)_2CH_2CH(CH_3)CH_3$

☐ **D** $CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$

(Total for question = 1 mark)

Q12.

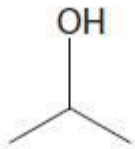
RocketRevise

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which compound has the **highest** boiling temperature?

www.RocketRevise.com

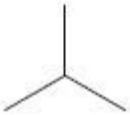
www.RocketRevise.com

☐ **A** 

☐ **B** 

☐ **C** 

RocketRevise

☐ **D** 

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

Q13.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which solvent dissolves the greatest amount of hydrocarbon $C_{35}H_{72}$?

- ☐ **A** butan-1-ol
- ☐ **B** ethanoic acid
- ☐ **C** hexane
- ☐ **D** water

(Total for question = 1 mark)

www.RocketRevise.com

www.RocketRevise.com

Q14.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is **not** correct about ice?

- ☒ **A** ice has a lower density than water
- ☒ **B** H_2O molecules are further apart in ice than in water
- ☒ **C** the H—O—H bond angle is the same in ice and in water
- ☒ **D** H_2O molecules in ice are held together by hydrogen bonds

RocketRevise

(Total for question = 1 mark)

Q15.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which intermolecular forces exist **between** the molecules of the compound shown?



- ☒ **A** hydrogen bonding and London forces only
- ☒ **B** hydrogen bonding and permanent dipole-permanent dipole forces only
- ☒ **C** London forces and permanent dipole-permanent dipole forces only
- ☒ **D** hydrogen bonding, permanent dipole-permanent dipole forces and London forces

(Total for question = 1 mark)

RocketRevise

Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which compound does **not** form hydrogen bonds between its molecules?

- ☒ **A** HCl
- ☒ **B** H_2O
- ☒ **C** HF
- ☒ **D** NH_3

www.RocketRevise.com

www.RocketRevise.com

(Total for question = 1 mark)

Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Why is the boiling temperature of hydrogen iodide higher than that of hydrogen bromide?

- ☒ **A** hydrogen iodide has stronger London forces than hydrogen bromide
- ☒ **B** hydrogen iodide has a larger permanent dipole than hydrogen bromide
- ☒ **C** hydrogen iodide has stronger hydrogen bonds than hydrogen bromide
- ☒ **D** the H—I bond is stronger than the H—Br bond

RocketRevise

(Total for question = 1 mark)

Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these compounds would be expected to have the **highest** boiling temperature?

- ☒ **A** $(CH_3)_3COH$
- ☒ **B** $CH_3CH_2CH_2CH_2OH$
- ☒ **C** $CH_3CH_2CH(CH_3)_2$
- ☒ **D** $CH_3CH_2CH_2CH_2CH_3$

Use this space for any rough working. Anything you write in this space will gain no credit.

RocketRevise

(Total for question = 1 mark)

Q19.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

www.RocketRevise.com

www.RocketRevise.com

This question is about alkanes.

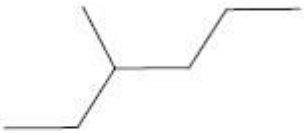
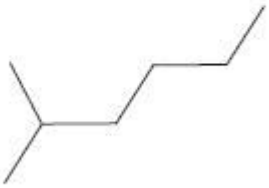
(a) Which of these alkanes has the highest boiling temperature?

(1)

- ☒ **A** butane
- ☒ **B** hexane
- ☒ **C** pentane
- ☒ **D** propane

(b) Which of these alkanes has the lowest boiling temperature?

(1)

- ☒ **A** 
- ☒ **B** 
- ☒ **C** 
- ☒ **D** 

RocketRevise

(Total for question = 2 marks)

Q20.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

RocketRevise

Which row in the table shows the forces between the molecules in the liquid state?

	Molecule	London forces	Permanent dipole-dipole interaction	Hydrogen bonding
☒ **A**	BF_3	✗	✓	✗
☒ **B**	CH_4	✓	✗	✓
☒ **C**	NH_3	✓	✗	✗
☒ **D**	H_2S	✓	✓	✗

www.RocketRevise.com

www.RocketRevise.com

(Total for question = 1 mark)

Q21.

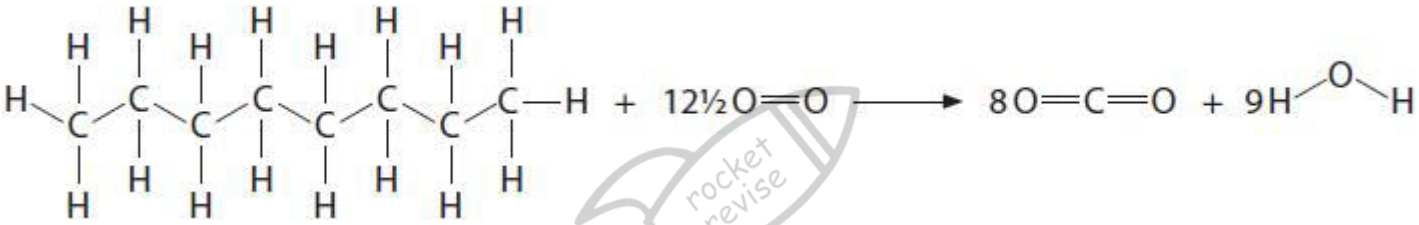
This question is about fuels.

The enthalpy change of combustion is the maximum amount of heat energy that can be obtained by the complete combustion of a substance.

Values for the enthalpy change of combustion may be obtained by direct measurement or from mean bond enthalpies. These data may be used to assess the efficiency of fuels.

Petrol is a mixture of a large number of different hydrocarbons containing between four and twelve carbon atoms. Octane, C_8H_{18}, is a typical component of petrol.

(a) The equation for the combustion of octane is shown.



(i) Use mean bond enthalpies to calculate a value for the enthalpy change of combustion of octane.

(4)

Bond	C—C	C—H	O—H	O=O	C=O
Mean bond enthalpy /kJ mol^{-1}	347	413	464	498	805

(ii) The standard enthalpy change of combustion, $\Delta_c H^{\ominus}$, of octane is −5470 kJ mol^{-1}.
Give **two** reasons why this value, measured under standard conditions, is different from the value obtained using bond enthalpy data.

(2)

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

(iii) When petrol is used to power a car, the energy available is less than the theoretical maximum. When one kilogram of petrol powers a car, the energy used to move the car is 11 MJ. Calculate the percentage of the maximum energy that is available to move a car, assuming that this fuel is pure octane.

Use $\Delta_c H^{\ominus} = -5470\,\text{kJ mol}^{-1}$.

(2)

RocketRevise

(iv) Give **two** reasons why the energy used to move the car is less than the theoretical maximum.

(2)

.....

.....

.....

.....

* (b) Two alternatives to hydrocarbon fuels are hydrogen and ammonia. Hydrogen burns to form water as the only product but hydrogen is extremely hard to liquefy. Ammonia is easily liquefied but it is toxic.

Compare and contrast the intermolecular forces involved when hydrogen and ammonia liquefy, by describing how these forces are formed.

(6)

RocketRevise

(Total for question = 16 marks)

Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The boiling temperature of hexane is 69°C and that of cyclohexane is 81°C.

The main reason that cyclohexane has a higher boiling temperature is that cyclohexane molecules

☒ **A** have more electrons

www.RocketRevise.com

www.RocketRevise.com

☒ **B** have strong permanent dipole-permanent dipole forces
☒ **C** have a larger surface area of contact
☒ **D** form hydrogen bonds

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

RocketRevise

Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Hydrogen bonds are formed when methanol dissolves in water.

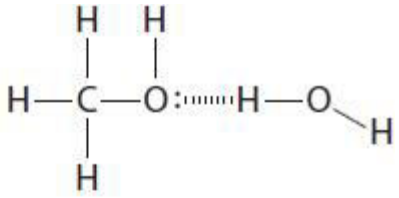
Which structure best represents a hydrogen bond between methanol and water?

RocketRevise

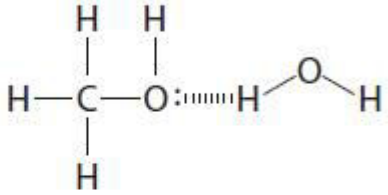
www.RocketRevise.com

www.RocketRevise.com

☐ **A**

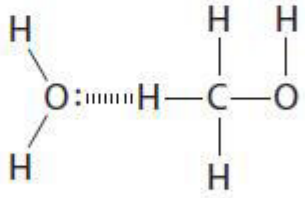


☐ **B**

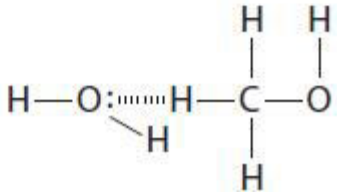


RocketRevise

☐ **C**



☐ **D**



(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

RocketRevise

Q24.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

www.RocketRevise.com

www.RocketRevise.com

about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which compound has intermolecular hydrogen bonding?

- ☐ **A** $(CH_3)_3N$
- ☐ **B** $(CH_3)_3CF$
- ☐ **C** $(CH_3)_3COH$
- ☐ **D** $(CH_3)_3CCHO$

(Total for question = 1 mark)

RocketRevise

Q25.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which compound has London forces as the **only** intermolecular force?

- ☐ **A** HF
- ☐ **B** OF_2
- ☐ **C** PF_3
- ☐ **D** CF_4

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

RocketRevise

Q26.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these isoelectronic compounds would be expected to have the **highest** boiling temperature?

- ☐ **A** butan–1–ol
- ☐ **B** 2–methylpropan–1–ol
- ☐ **C** 2–methylpropan–2–ol
- ☐ **D** pentane

www.RocketRevise.com

www.RocketRevise.com

(Total for question = 1 mark)

Q27.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which sequence shows the hydrogen halides in order of **decreasing** boiling temperature?

- ☐ **A** HF > HCl > HBr > HI
- ☐ **B** HF > HI > HBr > HCl
- ☐ **C** HI > HF > HCl > HBr
- ☐ **D** HI > HBr > HCl > HF

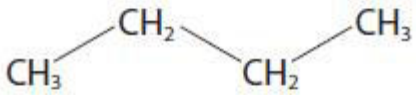
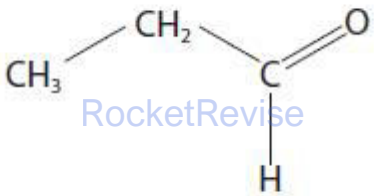
RocketRevise

(Total for question = 1 mark)

Q28.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structures and boiling temperatures of butane and propanal are shown.

Butane	Propanal
	
$T_b = 0\,°C$	$T_b = 49\,°C$

The main reason that propanal has a higher boiling temperature is that propanal molecules

- ☐ **A** have more electrons
- ☐ **B** have strong permanent dipole-permanent dipole forces
- ☐ **C** have a larger surface area of contact
- ☐ **D** form hydrogen bonds

www.RocketRevise.com

www.RocketRevise.com

(Total for question = 1 mark)

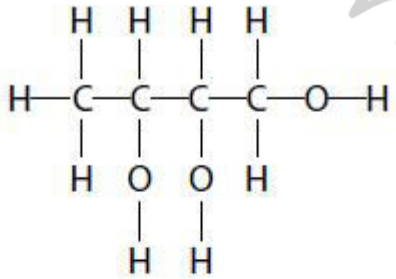
Use this space for any rough working. Anything you write in this space will gain no credit.

RocketRevise

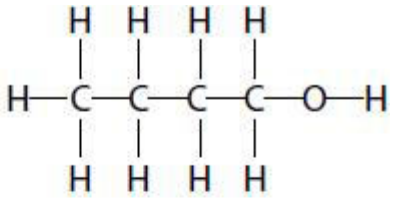
Q29.

Which is the order of **increasing** boiling temperature?

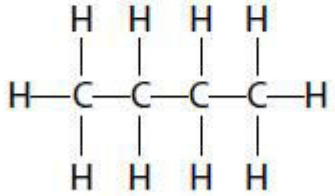
W =



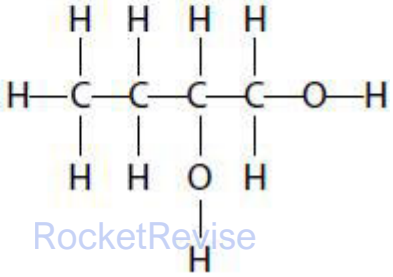
X =



Y =



Z =



RocketRevise

(1)

- ☒ **A** Y Z X W
- ☒ **B** Y X Z W
- ☒ **C** W X Z Y
- ☒ **D** X W Y Z

(Total for question = 1 mark)

Q30.

www.RocketRevise.com

www.RocketRevise.com

Which alkane has the highest boiling temperature?

(1)

- ☒ **A** 2,2-dimethylpropane
- ☒ **B** 2-methylbutane
- ☒ **C** butane
- ☒ **D** pentane

(Total for question = 1 mark)

RocketRevise

Q31.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which alkane has the strongest London forces in the liquid phase?

- ☒ **A** $(CH_3)_4C$
- ☒ **B** $CH_3CH_2CH_2CH_2CH_3$
- ☒ **C** $(C_2H_5)_4C$
- ☒ **D** $CH_3(CH_2)_7CH_3$

(Total for question = 1 mark)

Q32.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about boiling temperatures.

RocketRevise

(a) Which isomer has the **highest** boiling temperature?

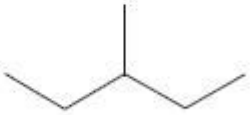
(1)

www.RocketRevise.com

www.RocketRevise.com

☒ **A** 

☒ **B** 

☒ **C** 

☒ **D** 

RocketRevise

(b) Which compound has the **highest** boiling temperature?

(1)

☒ **A** pentane
☒ **B** cyclopentane
☒ **C** butan-1-ol
☒ **D** butane-1,4-diol

(Total for question = 2 marks)

Q33.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which alkane has the **highest** boiling temperature?

☒ **A** hexane
☒ **B** pentane
☒ **C** 2-methylpentane
☒ **D** 2,3-dimethylbutane

RocketRevise

(Total for question = 1 mark)

Q34.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

www.RocketRevise.com

www.RocketRevise.com

about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which compound has intermolecular hydrogen bonding?

- ☐ **A** HF
- ☐ **B** H_2Te
- ☐ **C** AsH_3
- ☐ **D** SnH_4

(Total for question = 1 mark)

RocketRevise

Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A compound contains

- molecules with non-polar bonds
- permanent dipole-permanent dipole forces between its molecules.

What could be the formula of this compound?

Use the Data Booklet as a source of information.

- ☐ **A** NO
- ☐ **B** $BeCl_2$
- ☐ **C** PH_3
- ☐ **D** Cl_4

(Total for question = 1 mark)

RocketRevise

Q36.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In which process are intermolecular hydrogen bonds broken?

- ☐ **A** $H_2(g) \rightarrow 2H(g)$
- ☐ **B** $H_2O(l) \rightarrow H_2O(g)$
- ☐ **C** $H_2(l) \rightarrow H_2(g)$
- ☐ **D** $H_2(g) + ½O_2(g) \rightarrow H_2O(g)$

(Total for question = 1 mark)

www.RocketRevise.com

www.RocketRevise.com

Q37.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In which liquid solvent would the solubility of lithium chloride (LiCl) be lowest?

- ☐ **A** ammonia
- ☐ **B** cyclohexane
- ☐ **C** methanol
- ☐ **D** water

RocketRevise

(Total for question = 1 mark)

Q38.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Sodium chloride dissolves in water to form a solution because of the interactions between the sodium chloride and water. What are these interactions?

- ☐ **A** dipole-dipole
- ☐ **B** hydrogen bonds
- ☐ **C** ion-dipole
- ☐ **D** London forces

(Total for question = 1 mark)

RocketRevise

Q39.

Which is the best solvent for chloroethane?

- ☐ **A** ethanol
- ☐ **B** hexane
- ☐ **C** hydrochloric acid
- ☐ **D** water

(Total for question = 1 mark)

www.RocketRevise.com

www.RocketRevise.com

Q40.

Water is a molecule that is essential for life and it has some unusual properties.

(a) Draw a diagram to show hydrogen bonding between two H_2O molecules in ice.
Include bond angles and relevant dipoles and lone pairs in your answer.

(3)

RocketRevise

(b) Compare and contrast the effect of intermolecular forces on the properties of water and ammonia, using the data shown.

(3)

Molecule	M_r	Boiling temperature /K	Density at 223 K /kg m^{-3}	Density at 278 K /kg m^{-3}
NH_3	16.0	240	698	0.763
H_2O	18.0	373	926	1000

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 6 marks)

Q41.

The molecular masses and boiling temperatures of HF, HCl and HI are shown.

www.RocketRevise.com

www.RocketRevise.com

Compound	M_r	Boiling temperature / °C
HF	20.0	19.5
HCl	36.5	−85.0
HI	127.9	−35.1

(a) Which is the best estimate for the boiling temperature of HBr?

(1)

- ☒ **A** 59°C
- ☒ **B** −20°C
- ☒ **C** −67°C
- ☒ **D** −90°C

RocketRevise

(b) Why does HF not fit the pattern of the boiling temperatures of the other hydrogen halides?

(1)

- ☒ **A** it forms hydrogen bonds
- ☒ **B** it is much smaller than the other hydrogen halides
- ☒ **C** the other halogens are all electronegative
- ☒ **D** fluorine is the most reactive

(Total for question = 2 marks)

Q42.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which amine, with molecular formula C_3H_9N, has the **lowest** boiling temperature?

- ☒ **A** $(CH_3)_3N$
- ☒ **B** $CH_3CH_2NHCH_3$
- ☒ **C** $CH_3CH_2CH_2NH_2$
- ☒ **D** $CH_3CH(NH_2)CH_3$

RocketRevise

(Total for question = 1 mark)

www.RocketRevise.com